global AI bootcamp

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Speech Everything and (let them) Rise!

Dennie Declercq

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Pune Tech Community

Dennie Declercq

Microsoft MVP

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Dennie is Microsoft MVP Developer Technologies and has experience in accessibility with Microsoft technologies. In daily life Dennie is president and developer at DDSoft, a nonprofit that connects IT to People who are less tech-savvy. Dennie invented technical solutions and systems to help people with disabilities to participate in their daily life. Thanks to his autism he's the right man at the right spot to contribute as a volunteer in function of people with disabilities.

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Agenda

- A primer into Accessibility
- Speech for Accessibility
- Microsoft Cognitive Speech
- Speech for WPF
- Speech for Web
- Thanks // Q&A

A primer into Accessibility

People with disabilities

- Learning Disability
  - Reading
  - Understanding / Cognition
  - Structure
- Autism / Autism Spectrum Disorder
  - Structure
  - Confirmation
- Dyslexia
  - Reading
- Mental Health
  - Structure (sometimes)
  - Confirmation

Learning Disability?

Intellectual Functioning

- Also called intelligence
- Mental capacity
- IQ score between 70-75

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

- Social responsibility
- Self-esteem

Practical skills

- Activities daily living
- Personal care / healthcare

Oh yes…

- And the well known Accessibility too
  - Blindness/ Vision loss
  - Auditory disability
  - Motoric restrictions

The Persona Spectrum

	Permanent	Temporary	Situational
Touch	One arm	Arm injury	New parent
See	Blind	Cataract	Distracted driver
Hear	Deaf	Ear infection	Bartender
Speak	Non-verbal	Laryngitis	Heavy Accent

Speech for Accessibility

What is computer speech?

- Official name of technology: **Speech Synthesis**
- **Speech synthesis** is the artificial production of human speech. A computer system used for this purpose is called a **speech computer** or **speech synthesizer**, and can be implemented in software or hardware products. A **text-to-speech** (**TTS**) system converts normal language text into speech; other systems render symbolic linguistic representations like phonetic transcriptions into speech.[1] The reverse process is speech recognition.
- Source: Wikipedia: https://go.ddsoft.be/ttswiki

A brief history of computer speech

- The Mattel Intellivision game console offered the Intellivoice Voice Synthesis module in 1982.
- Also released in 1982, Software Automatic Mouth was the first commercial all-software voice synthesis program.
- Arguably, the first speech system integrated into an operating system was the 1400XL/1450XL personal computers designed by Atari, Inc. using the Votrax SC01 chip in 1983.
- The first speech system integrated into an operating system that shipped in quantity was Apple Computer's MacInTalk. The software was licensed from third-party developers Joseph Katz and Mark Barton (later, SoftVoice, Inc.) and was featured during the 1984 introduction of the Macintosh computer
- Android 1.6 has Text to Speech possibility (2009)
- Used in Alexa and as Software as a Service in AWS[58] (from 2017).

I am Belgian

- Lernout & Hauspie

Speech vs Read Aloud/ Voice Over

Read Aloud/ Voice Over

Blind/ Vision Loss

How does it work?

➔ Reads sequentially whole screen

(Separate) Speech

Learning Disability

Dyslexia

How does it work?

➔ Click a control and read aloud

(Separate) Speech use cases

- Dyslexia
- Learning Disabilities
- The need for confirmations
- (Public) transport

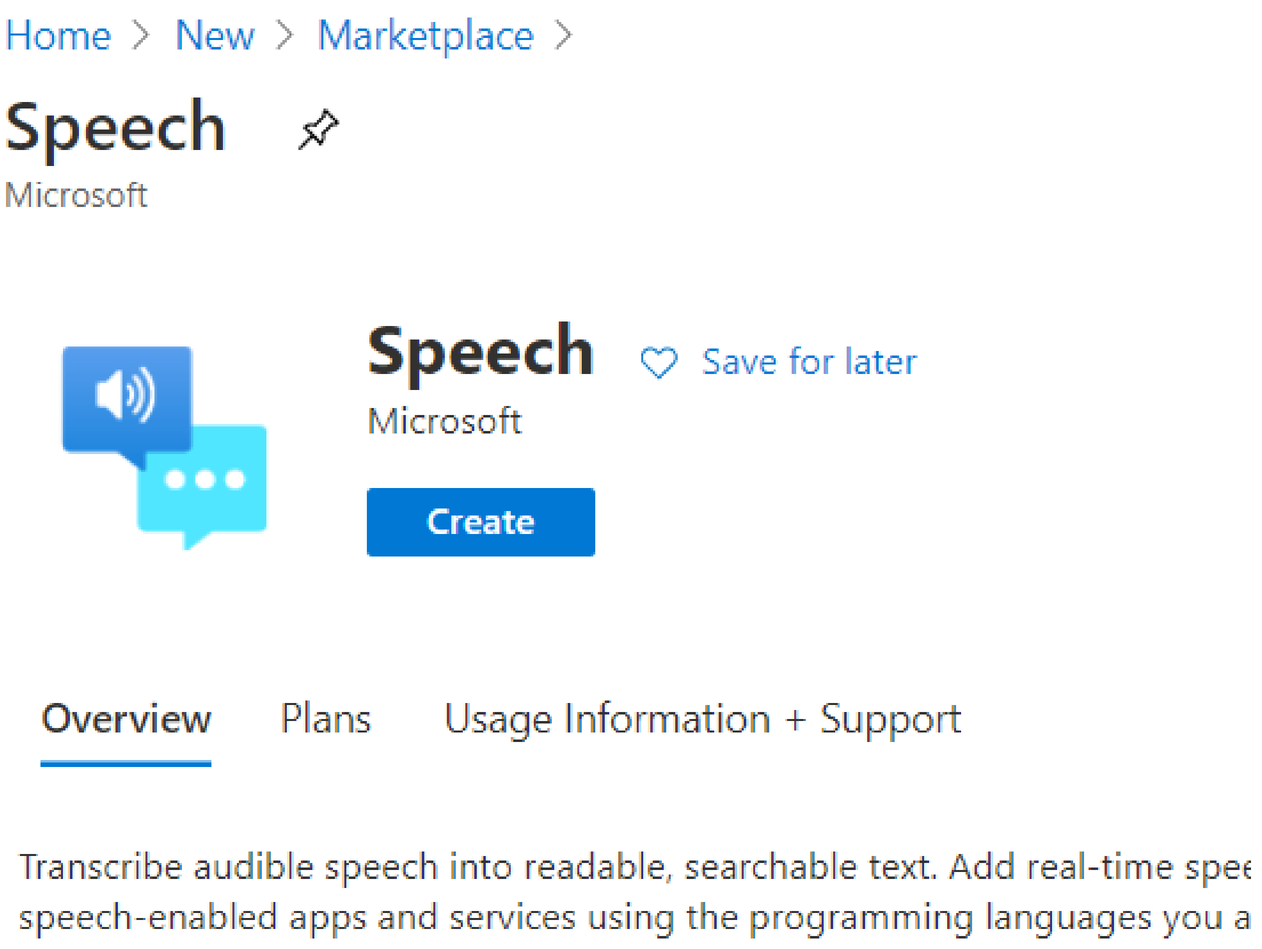
Microsoft Cognitive Speech

A part of Microsoft Cognitive Services

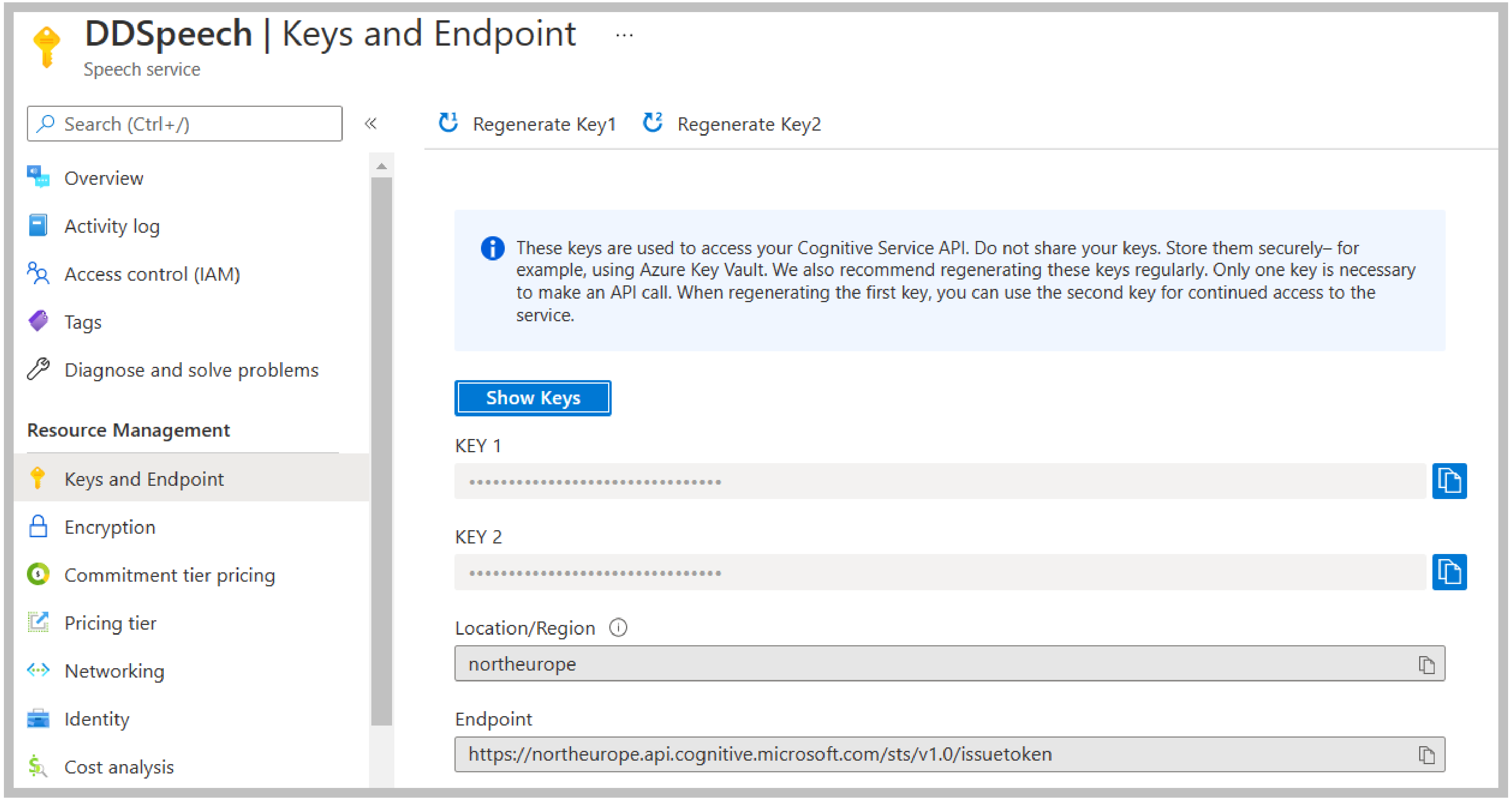
- Microsoft "developer" AI solution.
- Part of Azure
- Developer focused
- No need to be a math rockstar
- No need to be a science engineer

Microsoft
Cognitive Services

Making a Speech account



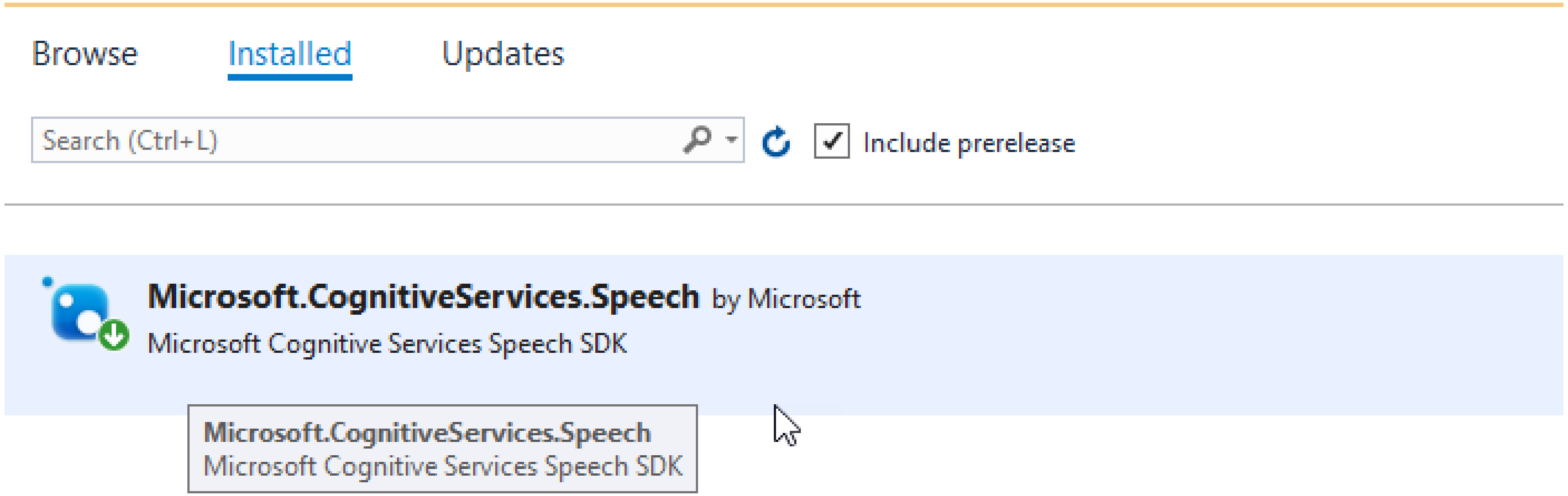
Getting your keys & endpoints



Demo

Making a Speech Service and accessing credentials

NUGET Package Speech SDK



Code to speak (SDK)

```
using Microsoft.CognitiveServices.Speech;
var _config = SpeechConfig.FromSubscription(KEY,REGION);
using (SpeechSynthesizer synth = new SpeechSynthesizer(_config))
    {
        await synth.SpeakTextAsync(text);
    }
```

REST: Making AccessToken

```
public async Task SetAutorizationTokenAsync(string fetchUri, string subscriptionKey)
    {
        HttpClient client = new HttpClient();
        client.DefaultRequestHeaders.Add("Ocp-Apim-Subscription-Key", subscriptionKey);
        var response = await client.PostAsync("https://northeurope.api.cognitive.microsoft.com/sts/v1.0/issueToken", null);
         _speech.AccessToken = await response.Content.ReadAsStringAsync();

    }
```

Accessing REST Speech Service


```
RESULT = STREAM GetTTS
string body = "<speak version='1.0' xml:lang='en-US'><voice xml:lang='" + _speech.Language + "' xml:gender='Male' name = 'en-US-ChristopherNeural'>" + text + " </voice></speak>";


        HttpClient client = new HttpClient();
        HttpContent content = new StringContent(body);
        content.Headers.ContentType = new System.Net.Http.Headers.MediaTypeHeaderValue("application/ssml+xml");
        client.DefaultRequestHeaders.Add("X-Microsoft-OutputFormat", "riff-16khz-16bit-mono-pcm");
        client.DefaultRequestHeaders.Authorization = new System.Net.Http.Headers.AuthenticationHeaderValue("Bearer", accessToken);
        client.DefaultRequestHeaders.Add("User-Agent", "TextHelp");
        HttpResponseMessage response = await client.PostAsync(_endpoint, content);
```

Saving AudioStream to .mp3 file


```
var readStream = await GetTTS(_speech.AccessToken, text);

         FileStream writeStream = File.Create(path);
         int Length = 256;
         Byte[] buffer = new Byte[Length];

         int bytestoRead = readStream.Read(buffer, 0, Length);
         while (bytestoRead > 0)
         {
             writeStream.Write(buffer, 0, bytestoRead);
             bytestoRead = readStream.Read(buffer, 0, Length);
         }

         readStream.Close();
         writeStream.Close();
```

API style vs SDK style

API
- Save audio to file
- Need to develop the file generation code
- Everywhere possible
- NO Send audio directly to speakers

SDK
- Send audio directly to Speakers
- Save audio to file
- Easy to use
- Few lines of code
- Not everywhere possible

Things to keep in mind

- Character Count
- Pause & more
- The pay model is via used characters

Demo

SPEECH via SDK

Demo

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SPEECH via web API

Speech for WPF

Accessing The Speech SDK thru WPF

- Why?
  - Make windows applications accessible for people with learning disabilities.
  - Use a read function on fields for to check written tekst for people with dyslexia
- What
  - Each control is possible
  - The help page
  - Nessecary elements

Demo

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SPEECH for WPF

Speech for Web

Speech on Hosted Web Apps

- SDK Speech can be added on ASP.NET hosted websites.
- This provides a whole new way of web accessibility for people with learning disabilities.
- This is the future if we want to include everyone.

Speech on WASM

- On Blazor running on Web Assembly you can only use REST Speech.
- Generate and save audio files to:
  - Web project (Azure Static Websites)
  - Azure Blob Storage

Demo

SPEECH The future of Accessible websites with Blazor

Q&A

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Thank You!